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MILITARY SPECIFICATION

REPAIR KIT, GLASS REINFORCED PLASTIC LAMINATE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a repair kit containing epoxy type thermosetting resins, in paste and liquid forms, with associate hardener, and glass reinforcing materials. This kit is used for making repairs to sections of reinforced plastic boats and other structures.

1.2 Classification. Repair kits shall be of the following types, as specified (see 6.2):

Type I	–	Standard repair kit
Type II	–	Refill kit
Type III	–	Special Kit

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-D-226	Depressor, Tongue, Wood
H-B-292	Brush, Adhesive, (Glue)
L-P-375	Plastic Film, Flexible, Vinyl Chloride
PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-C-186	Containers, Packaging, and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-C-795	Cushioning Material, Packaging (Flexible Cellular, Plastic Film) for Packaging Applications
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
UU-P-268	Paper, Kraft, Wrapping

MILITARY

MIL-Y-1140	Yarn, Cord, Sleeving, Cloth and Tape-Glass
MIL-P-17549	Plastic Laminates, Fibrous Glass Reinforced, Marine Structural
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-C-19663	Cloth, Woven Roving, for Plastic Laminate
MIL-M-43248	Mats, Reinforcing, Glass Fiber

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data Sheets, Transportation Data and Disposal Data for Hazardous Material Furnished to Government Activities
FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 613 Standard Test Method for Resin Content of Carbon and Graphite Prepregs by Solvent Extraction; (DOD adopted)

D 217 Standard Test Methods for Cone Penetration of Lubricating Grease; (DOD adopted)

D 638 Standard Test Methods for Tensile Properties of Plastics; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries of other information services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Materials.** Materials shall be free from contaminants and foreign or extraneous matter which may affect the performance characteristics of the resins or reinforcing materials as determined by first article test (see 4.3).

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3.3 Characteristics.

3.3.1 Contents. The contents of the kits shall be as indicated below.

3.3.1.1 Type I kit.

Item	Unit	Reference
1. Resin liquid	3 each 24 fluid ounces in 1-quart can	3.3.2
2. Resin liquid	3 each 12 fluid ounces in 1-pint can	3.3.2
3. Hardener, liquid	3 each 1/2-pint container	3.3.3
4. Hardener, liquid	3 each 1/4-pint container	3.3.3
5. Resin, paste	5 each 60 cubic centimeters (CM^3) in collapsible tubes	3.3.5
6. Hardener, paste	5 each collapsible tubes	3.3.6
7. Glass cloth	3 square yards	3.3.8.1
8. Woven roving cloth	5 square yards	3.3.8.2
9. Glass mat	6 square feet	3.3.8.3
10. Polyethylene gloves	6 pairs (3 pairs medium size 7-9 and 3 pairs large size 9-11)	3.3.10
11. Separating film	4 square yards	3.3.9
12. Eye protection	1 each	3.3.10
13. Wooden spatula (small)	10 each	3.3.11
14. Wooden stirring rods	5 each	3.3.12
15. Stirring spatula	5 each	3.3.12
16. Spreading tool	1 each	3.3.13
17. Brush, glue	4 each	3.3.14
18. Paper, kraft wrapping	3 square yards	3.3.15
19. Sandpaper, aluminum oxide, 80 grit	2 sheets	3.3.16
20. Scissors	1 each	3.3.16
21. Instruction publication	1 copy	3.3.17

3.3.1.2 Type II kit. The type II kit shall contain the same items as type I down through item 10, except the paste resin and hardener shall be reduced to two tubes each.

3.3.1.3 Type III kit. The type III kit shall contain only those materials, items, and quantities selected by the using activity (see 6.2). It is expected that this kit will be ordered with resin, hardener, and one or more types of fiberglass reinforcement. Any other items or fewer items may be included. However, the quantity of hardener supplied shall be proper to fully react the quantity of resin supplied to provide conformance to 3.3.4 and 3.3.7.4, as appropriate, when mixed one container of hardener to one container of resin, as supplied.

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3.3.2 **Liquid resin.** The liquid resin shall be of a thermosetting epoxy type, compounded with or without additional filler and shall be as specified in 3.3.2.1 through 3.3.2.3.

3.3.2.1 **Viscosity.** The liquid resin shall have a viscosity of not less than 5,000 centipoises and not more than 25,000 centipoises at 23 degrees Celsius (°C) (see 4.6.4.1).

3.3.2.2 **Stability.** The liquid resin shall be stable for a minimum period of 12 months (see 4.6.4.2).

3.3.2.3 **Flash point.** The liquid resin shall have a flash point greater than 121 °C (250 degrees Fahrenheit (°F)) as determined by the Cleveland open cup flash point method specified in test method 1103.7 of FED-STD-791.

3.3.3 **Liquid hardener.** The liquid hardener shall be a liquid that shall mix easily with the resin, and shall conform to 3.3.3.1 through 3.3.3.3.

3.3.3.1 **Viscosity.** The liquid hardener shall have a viscosity of not more than 25,000 centipoises at 23°C (see 4.6.5.1).

3.3.3.2 **Stability.** The liquid hardener shall be stable for a minimum period of 12 months (see 4.6.5.2).

3.3.3.3 **Flash point.** The liquid hardener shall have a flash point greater than 93.3 °C (200 °F) as determined by the Cleveland open cup flash point method specified in test method 1103.7 of FED-STD-791.

3.3.4 **Performance requirements for liquid resin and hardener.** The two-component system consisting of liquid resin and hardener, when mixed one can of resin to one can of hardener in the proportions supplied in the containers, shall have the following characteristics:

a. *Viscosity.* The liquid resin and hardener shall have a viscosity of not less than 5,000 centipoises and not more than 18,000 centipoises (see 4.6.6.1).

b. *Reaction time.* The liquid resin and hardener shall react in not less than 25 minutes and not more than 60 minutes after combining (see 4.6.6.2). No external heat shall be applied for initiating the reaction or for curing.

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c. *Test patches.* The liquid resin and hardener, when used to make test patches (see 4.6.3) combined with the reinforcing fibrous glass materials, and when applied, shall produce patch specimens having the following physical properties (see 4.6.3):

Average flexural breaking load	Dry, 25 pounds, minimum
	Wet, 25 pounds, minimum
Average tensile breaking load	Dry, 860 pounds, minimum

d. *Filler.* Filler, where used in the liquid resin-hardener formulation, shall be inorganic and shall not exceed 8 percent by weight (see 4.6.2).

3.3.5 **Paste resin.** The paste resin shall be of a thermosetting epoxy type compounded with filler material. It shall mix easily with the hardener, and shall be as specified in 3.3.5.1 through 3.3.5.4.

3.3.5.1 **Consistency.** The paste resin shall have a penetration value of 300 to 180 (in 1/10 millimeter units) (see 4.6.8.1).

3.3.5.2 **Color.** The paste resin shall be white in color, so that when mixed with the black hardener, the mixture will blend into a grey color to indicate that the materials have been homogeneously mixed.

3.3.5.3 **Stability.** The paste resin shall be stable for a minimum period of 12 months (see 4.6.8.2).

3.3.5.4 **Quantity.** The paste resin shall be furnished in units of 60 $\pm$ 10 cm^3 (approximately 2 fluid ounces), packaged in a collapsible metal or plastic tube with a screw top cap.

3.3.6 **Paste hardener.** The paste hardener shall mix easily with the paste resin, react as specified in 3.3.7, and shall be as specified in 3.3.6.1 through 3.3.6.4.

3.3.6.1 **Consistency.** The paste hardener shall have a penetration value of 300 to 180 (in 1/10 millimeter units) (see 4.6.8.1).

3.3.6.2 **Color.** The paste hardener shall be black in color, so that when properly mixed with the white paste resin, the mixture will blend into a grey color to indicate that the materials have been homogeneously mixed.

3.3.6.3 **Stability.** The paste hardener shall be stable for a minimum period of 12 months (see 4.6.8.2).

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3.3.6.4 **Quantity.** The quantity supplied shall react with the 60 cm^3 of paste resin supplied in each resin tube, and the resultant mixture shall conform to 3.3.7. The paste hardener shall be packaged in a collapsible metal or plastic tube with a screw cap. The size of the tube opening shall provide that the volume of hardener extruded from the tube in a unit length shall be the amount required to properly react with the quantity of resin extruded from the resin tube in an equivalent unit length.

3.3.7 **Performance requirements for paste resin and hardener.** The paste resin and hardener, when mixed in the proportions supplied in the collapsible tubes, shall conform to 3.3.7.1 through 3.3.7.4.

3.3.7.1 **Reaction time.** The paste resin and hardener, when mixed (see 4.6.9.1), shall react (harden) in not less than 20 minutes and not more than 50 minutes after combining. No external heat shall be applied for initiating the reaction or for curing.

3.3.7.2 **Ratio.** The ratio of paste hardener to paste resin that is necessary to obtain a cure shall be constant and as supplied in the collapsible metal tubes. The sizes of the openings of the tubes of resin and hardener shall permit the proper ratio of resin and hardener to be expelled when equal lengths of paste material are used.

3.3.7.3 **Flow.** The paste resin and hardener, when mixed and applied (see 4.6.9.2), shall not flow.

3.3.7.4 **Test patches.** The paste resin and hardener, when used to make test patches (see 4.6.7) and when applied, shall produce patch specimens having the following physical properties (see 4.6.7.4 and 4.6.7.5).

a.	Average flexural breaking strength, dry	4,000 pounds per square inch (lb/in^2)
b.	Average tensile breaking strength, dry	2,500 lb/in^2

3.3.8 **Glass reinforcements.** Glass reinforcements shall be of the following types and include a finish on the fiber (such as VOLAN A) so that they are compatible with the epoxy resin provided as specified herein. The reinforcement system shall provide for good wetting of the fiber and high wet strength retention of the laminated patch.

3.3.8.1 **Glass fiber cloth.** The glass fiber cloth shall conform to form 4, class C, type 1000-150 of MIL-Y-1140. The material shall be supplied in, at most, two pieces whose total area shall be at least 3 square yards. No single piece shall be less than 1 yard square.

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3.3.8.2 **Woven roving.** Woven roving shall conform to MIL-C-19663, of nominal 24 ounces per square yard. Material shall be in two pieces at most, totalling 5 square yards, and no piece shall be smaller than 1 yard square.

3.3.8.3 **Fibrous glass mat.** The fibrous glass mat shall conform to type I, class 1 of military specification MIL-M-43248, 1-1/2 ounces per square foot. Six square feet minimum in one piece shall be provided.

3.3.9 **Separating film.** The separating film shall consist of one piece of vinyl film at least 4 square yards in area and not less than 1 yard wide. The material shall conform to type I, class 1 of L-P-375, 0.0040 $\pm$ 0.0004 inch thick.

3.3.10 **Gloves and eye protection.** Gloves shall be polyethylene having a minimum thickness of 1-1/2 mils. Eye protection shall be an eye shield, glasses or goggles, of commercial quality and suitable for protection of the eyes from splash or splatter or kit materials.

3.3.11 **Spatula, small.** The small wooden spatula shall be a smooth hardwood tongue depressor, approximately 5-3/4 inches long, 11/16 inch wide, conforming to GG-D-226.

3.3.12 **Stirring tool spatula.** The stirring tool shall be of smooth hardwood, approximately 9 inches long, 3/16 inch thick, and 1-1/8 inches wide, with shaped handle. The wooden stirring rods shall be 1/4 inch in diameter and 6 $\pm$ 1/2 inches long.

3.3.13 **Spreading tool.** The spreading tool shall consist of a kitchen type spatula with a rubber or polyethylene spreader. The spreader shall be approximately 2 by 3-1/2 inches. The wooden handle shall be smooth and shall extend approximately 6 inches beyond the spreader. The length shall be 9-1/2 $\pm$ 1/2 inches overall.

3.3.14 **Brush, glue.** The brush shall conform to type F, size 1-inch of H-B-292.

3.3.15 **Paper, kraft.** The kraft paper supplied shall consist of one piece, 3 square yards in size, of 30 pounds basis weight minimum, conforming to UU-P-268.

3.3.16 **Scissors and sandpaper.** Two sheets of aluminum oxide sandpaper, 80 grit, shall be supplied along with one pair of metal bladed scissors that shall easily cut the woven roving fabric supplied with the kit.

3.3.17 **Instructions.** One copy of "Instructions for Repair Kits of Reinforced Plastics", sealed in a waterproof envelope, shall be furnished with each repair kit. The instructions shall be as specified in the appendix to this specification, except that additional safety and safe handling procedures shall be included as required for the specific products involved.

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3.4 **Toxicity.** The components of this kit and the final product after the components are mixed together shall have no adverse effect on the health of personnel when the components and product are used with adequate ventilation, under normal conditions, and for the intended purpose. Questions pertaining to toxicity shall be referred to the Naval Medical Command (NAVMEDCOM), who will act as an advisor to the contracting activity.

3.5 **Special markings.**

3.5.1 **Labeling for liquid resin container.** The following information shall be provided on a durable printed label:

"WARNING: MAY CAUSE SKIN
IRRITATION. VAPOR HARMFUL.
(Add additional warnings/cautions as necessary)

'A' – Liquid Epoxy Resin for Repair Kit-Contents (Nominal)
(1 quart or 1 pint).
Packaged by (manufacturer's name and address).
Date of manufacture (month and year).
Lot number.
Store in a cool place.

CAUTION

Do not mix hardener and resin together until actually ready for use.

CAUTION

Use only with liquid hardener provided in this kit."

The following instruction for mixing of the resin and hardener shall be detailed on the label:

"Instructions for Mixing Liquid Resin for Use

1. Before mixing this resin with hardener, read instruction sheets provided with kit.
2. Before mixing this resin, the preparation of the area and tailoring of glass reinforcement should be finished.
3. Remove the cover of container 'A' (resin). Open the proper size container 'B' (hardener) and put its entire contents into container 'A'
   while stirring the resin.
   MIX THOROUGHLY.

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NOTE

The resin (container 'A') after mixing with hardener (container 'B') has a working life of about 1/2 hour at 73 °F. Working life will be much shorter at higher temperatures and much longer at lower temperatures. Cool or warm the closed resin container in cold or hot water where necessary before mixing. Allow no water to come in contact with the resin."

3.5.2 **Labeling for liquid hardener container.** The following information shall be provided:

"'B' – Liquid Hardener for Epoxy Repair Kit.
Packaged by (manufacturer's name and address).
Date of manufacture (month and year).
Lot number.
Store in a cool place.

WARNING: MAY CAUSE SKIN
IRRITATION. VAPOR HARMFUL.
(Add additional warnings/cautions as necessary)

Avoid exposure to vapor. Use respirator if adequate ventilation is not available. Avoid contact with the skin or eyes. Use protective clothing, gloves, and goggles. In case of accidental contact, promptly flush skin or eyes with plenty of water for 15 minutes; get medical attention for eyes.

CAUTION

Use only with liquid epoxy resin provided in this kit."

3.5.2.1 **Marking for larger container.** In addition to the above markings, the larger (1/2-pint size) liquid hardener containers shall be marked as follows:

"The entire contents of this container of hardener shall be mixed with the liquid resin in the 1-quart size resin can."

3.5.2.2 **Marking for smaller container.** The smaller (1/4-pint size) liquid hardener container shall be marked as follows:

"The entire contents of this container of hardener shall be mixed with the liquid resin in the 1-pint size resin can."

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3.5.3 **Labeling for paste resin tube.** The following information shall be provided on each tube of resin. This information shall be printed on the tube or on a durable printed label attached to the tube:

"Paste Epoxy Resin for Repair Kit.
Packaged by (manufacturer's name and address).
Date of manufacture (month and year).
Lot number.
Store in a cool place.

WARNING: MAY CAUSE SKIN IRRITATION. AVOID SKIN CONTACT
(Add additional cautions/warnings as necessary)

CAUTION

Use only with paste hardener provided in this kit."

3.5.3.1 The following information shall be provided on each resin tube:

"For repair of minor damage, see instruction sheet.
Use flat clean surface for mixing.
Squeeze equal lengths of resin and hardener.
MIX THOROUGHLY.
Apply to clean surface.
At 73 °F, will solidify in 1/2 hour; will cure (harden) in 24 hours."

3.5.4 **Labeling for paste hardener container.** The following information shall be provided on the tube of hardener:

"Paste Hardener for Epoxy Repair Kit.
Packaged by (manufacturer's name and address).
Date of manufacture (month and year).
Lot number.

WARNING: MAY CAUSE IRRITATION. AVOID CONTACT WITH SKIN.

If contact occurs, wash immediately with soap and water.

CAUTION

Use only with paste epoxy provided in this kit."

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3.5.5 **Labeling for paste resin-hardener package.** Each package shall be marked with the following information:

"This package contains the following repair items:

1 tube of paste resin.
1 tube of paste hardener.
2 mixing sticks.

Read instructions before using.
Packed by (manufacturer's name and address).
Date of manufacture (month and year).
Store in a cool place."

3.5.6 **Marking of kit.** Each kit shall be marked with the following information:

"Kit number (number in numerical sequence).
Contract or order number.
For repair of reinforced plastic structures.
Packed by (manufacturer's name and address).
Store in cool place.
Storage life at 125 °F, 12 months.
Date of manufacture (month and year)."

3.6 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.7 **Material safety data sheet (MSDS).** The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 5.5. and 6.6).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on one complete kit and an additional quart of paste resin with the proper amount of paste hardener. Examination and tests shall be as specified in 4.5 and 4.6.

4.4 Quality conformance inspection.

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For purposes of sampling, a lot shall consist of not more than 200 kits of the same type, having resins and hardeners from the same batches, respectively, and offered for delivery at the same time.

4.4.1.2 Sampling for visual and dimensional examination. A random sample of repair kits shall be selected from each lot offered for examination of visual and dimensional characteristics (see 4.5.1 and 4.5.1.1). Sampling shall be in accordance with MIL-STD-105, general inspection level I.

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4.4.1.3 **Sampling for lot acceptance tests.** For lot acceptance tests, two repair kits shall be selected at random from each lot. One of the two sample kits shall be used to provide material for and be subjected to the tests of 4.6.3.

4.5 **Examination and test.**

4.5.1 **Visual and dimensional examination.** Each of the sample kits selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification not involving tests. Any kit in the sample containing one or more visual or dimensional defects shall be rejected.

4.5.1.1 **Filled containers.** The filled containers shall be removed from each of the sample kits and examined for conformance to the applicable requirements of this specification.

4.5.2 **Lot acceptance tests.** Each sample kit shall be subjected to the tests specified in 4.6.2, 4.6.4.1, 4.6.5.1, 4.6.6.1, 4.6.6.2, 4.6.8.1, 4.6.9.1, and 4.6.9.2. If either of the sample kits selected fails any of the tests, the lot shall be rejected.

4.6 **Test procedures.**

4.6.1 **Conditioning.** Unless otherwise specified herein, tests shall be performed at a room temperature of 23 ± 1 °C and a relative humidity of 50 ± 4 percent.

4.6.2 **Percentage filler.** The percentage filler in the liquid resin-hardener system shall be determined by ASTM C 613. The residue after ignition shall be reported as filler and shall meet 3.3.4(d).

4.6.3 **Test patches, liquid resin-hardener system.**

4.6.3.1 **Preparation of standard panel for patching.** The standard section to be patched shall be a 0.250 ± 0.015 inch thick fire-retardant polyester resin-glass cloth reinforced panel approximately 7 by 10 inches or longer in size conforming to grade 2 or 3 of MIL-P-17549. Along one 10-inch side of the panel a 1-inch wide scarf shall be cut (see figure 1). The scarf may be prepared by cutting on a saw or on a planer or shaper. After the scarf has been cut it shall be abraded with number 60 Garnet paper. Dust or grit remaining on the abraded scarfed area shall be removed by clean air blast. The prepared panel shall be placed on a separating film-covered flat wooden or metal supporting plate where the patch will be applied.

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4.6.3.2 **Application of standard patch using liquid resin-hardener system.** Sixteen plies of glass cloth type 1000-150, conforming to the requirements of 3.3.8, shall be cut to 8- by 10-inch sizes. One small can of the liquid resin supplied with the kit shall be mixed thoroughly with the amount of hardener supplied in the 1/4-pint liquid hardener container. The resin shall be used to impregnate the 16 plies of glass cloth by spreading, brushing, or in any other convenient manner. The patch may be applied to the standard panel by either of the following procedures:

a. The impregnated plies shall be laid up to form a 1-inch taper as shown on figure 2. Over the layup of impregnated plies of cloth a sheet of separating film shall be placed and the air bubbles shall be worked out. The film shall then be removed. Additional resin shall be applied over the scarfed area of the standard panel and the overlapping plies of the impregnated cloth (patch) shall be placed to fit the 1-inch scarf of the panel. Over the patched panel a sheet of separating film shall be placed.

b. The plies of glass cloth shall be laid up by placing one ply at a time against the standard taper. The glass shall be impregnated in place until all 16 plies are in position along the 1-inch scarf of the panel. Over the patched panel a sheet of separating film shall be placed and excessive air bubbles worked out.

After the patch has been applied as described in (a) or (b), a flat wooden or metal plate together with weights totalling not more than 30 pounds shall be placed over the patch to provide contact pressure. The patched panel shall be permitted to set and cure for 24 hours at standard laboratory conditioning before being cut into test specimens (see figure 3).

4.6.3.3 **Flexural specimen preparation.** The patch prepared as specified in 4.6.3.2 shall be cut in 1/2-inch wide specimens with the scarfed area at the center of the specimen and cut to lengths as shown on figure 4.

4.6.3.3.1 **Dry flexural breaking load.** Five specimens shall be held at standard laboratory conditions for 24 hours after preparation. The specimens shall then be tested in flexure with the load applied at the center of the scarfed area with the scarf up as shown on figure 5. The span shall be 8 inches. The speed of testing shall be 0.25 inch per minute. The dimensions of the specimen shall be measured at the point at which the load is applied. Thickness and width shall be measured to 0.001 inch. The maximum load at failure shall be noted. The flexural breaking load shall be considered as the breaking load adjusted to the value for a 0.500-inch wide specimen.

4.6.3.3.2 **Wet flexural breaking load.** Five specimens shall be immersed in boiling distilled water for 2 hours. After the 2-hour boil, the specimens shall be placed in distilled water at 23 ± 1 °C for 1 hour. The specimens shall then be removed and immediately tested for flexural strength as specified in 4.6.3.3.1.

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4.6.3.4 **Tensile breaking load of patch.** Five 1/2-inch wide specimens prepared as specified in 4.6.3.3 shall be tested for tensile strength in accordance with ASTM D 638, except that the specimens shall not be necked down. The breaking load in tensile shall be reported as the maximum load sustained adjusted to the value for a 0.500-inch specimen.

4.6.3.5 **Retest of patches.** If the patch specimens tested do not conform to 3.3.4, one additional patch shall be prepared in accordance with 4.6.3. If specimens from this second patch are shown to have an average strength as specified in 3.3.4, the resin and hardener shall be deemed to conform to this specification.

4.6.4 **Liquid resin.**

4.6.4.1 **Viscosity.** The viscosity of the resin specified in 3.3.2.1 shall be determined by Brookfield Synchro-lectric Viscosimeter using a number 3 spindle at a speed of 4 revolutions per minute (r/min).

4.6.4.2 **Stability of the resin.** Resin shall be stored in an airtight container as specified in 5.2.1(a) at a temperature of 125 °F for a period of 12 months, after which it shall be tested for compliance with 3.3.2.1 and 3.3.4. It shall show no separation or precipitation of filler material.

4.6.5 **Liquid hardener.**

4.6.5.1 **Viscosity of the hardener.** The viscosity of the hardener shall be determined with a Brookfield Synchro-lectric Viscosimeter, at a spindle speed of 4 r/min.

4.6.5.2 **Stability of the hardener.** The hardener shall be stored in an airtight container as specified in 5.2.1(b), for a period of 12 months at 125 °F, after which it shall be tested for compliance with 3.3.3.1 and 3.3.4.

4.6.6 **Liquid resin hardener mixture.**

4.6.6.1 **Viscosity test.** The entire quantity of liquid hardener in the small (1/4-pint size) container shall be added to the liquid resin in the 1-pint can supplied with the kit and stirred well for 2 minutes. Five minutes after addition of the hardener to the resin, the viscosity of the mixture shall be determined in the can by a Brookfield Synchro-lectric Viscosimeter (without a guard arm), using a number 3 spindle at a speed of 4 r/min.

4.6.6.2 **Reaction time test.** The entire quantity of liquid hardener in the small (1/4-pint size) container shall be added to the liquid resin in the 1-pint can supplied with the kit and thoroughly mixed. The container shall be placed on a 1/2-inch thick insulating pad. A stop watch shall be started at the time the hardener is poured into the resin. The resin-hardener mixture shall be

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stirred every 5 minutes. When stirring becomes exceedingly difficult due to the altering of the normal consistency, the mixture shall be considered to have reached the end of its working life and the time shall be recorded. The difference between this time and the time at which the hardener was added is the reaction time of the liquid resin-hardener mixture.

4.6.7 Test patches, paste resin-hardener system.

4.6.7.1 Preparation of standard panel for patching. The standard panel to be patched shall be as specified in 4.6.3.1.

4.6.7.2 Application of standard patch using paste resin. The standard 1/4-inch thick section (see 4.6.7.1) to be patched shall be fixed or clamped in place over a sheet of separating film. Two guide bars, 1/4-inch thick and extending 10 inches beyond the tapered edge, shall be placed along the 7-inch sides of the panel. Paste resin and hardener shall be mixed in the prescribed ratio, and the paste mixture shall be applied contiguous to the 1-inch taper of the standard panel to make a patch 8 by 10 inches. A sheet of separating film shall be placed over the patch, and the paste material between the guides shall be rolled to form a patch 1/4-inch thick. The patched panel shall be permitted to cure for 24 hours at standard laboratory conditioning before being cut into test specimens.

4.6.7.3 Flexural specimen preparation. The patch prepared as specified in 4.6.7.2 shall be cut in 1/2-inch wide specimens with the scarfed area at the center of the specimen and shall then be cut to length as shown on figure 4.

4.6.7.4 Dry flexural strength. Five specimens shall be conditioned at laboratory conditioning for 24 hours after preparation. The specimens shall be tested in flexure with the load applied at the center of the scarfed area with the scarf up as shown on figure 5. The span shall be 8 inches. The speed of testing shall be 0.25 inch per minute. The dimensions of the specimen shall be measured at the point at which the load is applied. Thickness and width shall be measured to 0.001 inch. The maximum load at failure shall be noted. The flexural strength shall be considered as the maximum fiber stress computed from the following equation:

$$S = \frac{3PL}{2bd^2}$$

Where:

P = maximum load sustained (pounds)
L = span = 8 inches
b = width (inches)
d = thickness (inches).

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4.6.7.5 **Tensile strength of patch.** Five 1/2-inch wide specimens prepared as specified in 4.6.3.3 shall be tested for tensile strength in accordance with ASTM D 638 except that the specimen shall not be necked down. The ultimate strength in tension shall be reported as the maximum load sustained divided by the cross-sectional area measured at the center of the joint.

4.6.7.6 **Retest of patches.** If the patch specimens tested do not conform to 3.3.7.4, one additional patch shall be prepared and tested as specified in 4.6.7. If specimens from this second patch are shown to have an average strength as specified in 3.3.7.4, the resin and hardener shall be deemed to conform to this specification.

4.6.8 ***Paste resin and paste hardener.***

4.6.8.1 **Consistency.** The consistency of the paste resin and paste hardener shall be determined by means of its ability to resist penetration of a weighed cone.

4.6.8.1.1 **Apparatus.** An apparatus shall be used which will permit the cone to penetrate without appreciable friction. The apparatus shall be calibrated accurately to read in tenths of a millimeter (mm) in accordance with ASTM D 217.

4.6.8.1.2 **Cone.** The cone shall be in accordance with ASTM D 217. The tip and face of the cone shall be highly polished. The cone assembly shall have a total weight of 52.5 grams. The cone and its plunger assembly shall have a total moving weight of 100 grams.

4.6.8.1.3 **Test specimen container.** The container for holding the material to be tested shall be a flat bottom cylindrical can approximately 55 mm (2-3/16 inches) in diameter and 35 mm (1-3/8 inches) deep.

4.6.8.1.4 **Procedure.** The test material shall be packed into the test specimen container until flush with the top. If desired, a straightedge may be drawn across the surface of the sample to remove any excess material and present a smooth surface. The container and test material shall be centered beneath the penetrometer cone and assembly. The apparatus shall be leveled, and the plunger lowered until the tip of the penetrometer cone just touches the sample surface (watching the shadow of the tip aids in accurately setting the cone). The scale shall then be adjusted so that the scale activating device is in contact with the top of the rod holding the penetrometer cone, and the scale reading shall be recorded. The plunger shall be released suddenly and kept released for 5 seconds. The scale-activating device shall be moved until it is again in contact with the top of the rod holding the penetrometer cone, and the scale reading recorded. The penetration in mm is the difference between the scale readings. Three tests shall be made and the average reported. The sample shall be smoothed over before each test. After each penetration, the cone shall be carefully cleaned to remove all adhering material.

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4.6.8.2 **Stability.** The paste resin or paste hardener shall be stored in its collapsible metal tube as specified in 5.2.1(c) or 5.2.1(d) for a period of 12 months at 125 °F. After a 12-month storage period has elapsed, samples shall be tested for compliance with 3.3.5.1 or 3.3.6.1 and 3.3.7.

4.6.9 **Paste resin hardener mixture.**

4.6.9.1 **Reaction time test (paste resin-hardener).** A total of 120 grams of paste resin and hardener, in the prescribed ratio and supplied in the collapsible tubes, shall be mixed in a 6-ounce paper cup with a wooden tongue depressor. The time at which the activator is added to the resin shall be recorded. The resin-hardener mixture shall be stirred every 5 minutes. Five complete revolutions shall be made within a total time of 5 seconds. When stirring becomes exceedingly difficult because of the altering of the normal consistency, the mixture shall be considered to have reached the end of its working life and the time shall be recorded. The difference between this time and the time at which the hardener was added is the reaction time of the paste resin-hardener mixture.

4.6.9.2 **Flow test.** A channel 5 inches long, 3 inches wide, and 1/16 inch deep shall be cut in a mild steel block, measuring 5 by 3-1/2 by 3/8 inches. A well-mixed sample of the paste resin and hardener shall be placed in the channel to more than completely fill it. The excess at the top and ends shall be removed by drawing a straightedge across the material. The panel so prepared shall be permitted to stand in a vertical position, supported at its edges so that the surface beneath the paste resin-hardener mixture remains clear. The panel shall be observed to determine whether there is any flow of the material below its lower edge by the time the material solidifies.

4.6.10 **Toxicological formulations.** The contractor shall have the toxicological formula information and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.7 **Inspection of packaging.** Sample packages and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

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5.1 General

5.1.1 Navy Shipboard stowage fire-retardant requirements.

a. *Treated lumber and plywood.* When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II – weather resistant
	–	Category 1 – general use
Level C	–	Type I – non-weather resistant
	–	Category 1 – general use.

b. *Fiberboard.* When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

5.2.1 Level A. Preservation of the repair kit components shall be as follows:

a. *Liquid resin.* Cans shall conform to type V, class 2 of PPP-C-96. Seaming compound shall not be used. Exterior plan A and B coating and side seam stripping shall be required. The multifriction plug closure shall be in accordance with the appendix to the can specification using spot soldering or, as an alternate, safety clips.

b. *Liquid hardener.* Bottles shall conform to group A, class 1 of PPP-C-186, except that the glass shall be clear. The bottle shall be closed with metal continuous thread screw caps. The caps shall be provided with a tight fitting, nonabsorbent lining material. Lining material shall not affect or be affected by the product packaged. After filling of containers, caps shall have torsion applied to seal containers against leakage during handling, shipment, and storage. The bottle shall be further sealed with a cellulose cap or equivalent, covering the metal screw cap externally and extending well down on the neck of the bottle. Each bottle shall be cushioned by wrapping with fire-retardant cushioning conforming to PPP-C-795, class 3 and individually placed in a snug-fitting folding or set-up paperboard or fiberboard container as specified in table I, selection and closure of unit intermediate containers, appendix F or MIL-STD-2073-1. Containers shall be of the weather or water resistant class, variety, or process as applicable. Container closure shall be in accordance with the applicable container specification or appendix thereto. Container selection shall be at the option of the contractor.

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c. *Paste resin.* The paste resin shall be unit packaged in metal or plastic collapsible tubes. The tubes shall be closed with a screw cap.

d. *Paste hardener.* The paste hardener shall be unit packaged in plastic or metal collapsible tubes. The tubes shall be closed with a screw cap. The tube material shall not affect or be affected by the product packaged.

e. *Paste resin and hardener.* One tube of paste resin, one tube of paste hardener, and two small wooden spatulas (tongue depressor type) shall be placed in a snug-fitting folding or set-up paperboard or fiberboard container as specified under (b) above.

f. *Glass reinforcements.* Glass reinforcements shall be individually unit packaged in a bag fabricated from 0.003 inch (minimum) thickness polyethylene material. Bag closure shall be by heat sealing.

g. *Separating film.* The separating film shall be folded to a minimum practicable size and secured by taping or tying.

h. *Gloves.* Gloves shall be unit packaged in a bag fabricated of neutral kraft material. Closure of the bag shall be effected by taping or with adhesives.

i. *Spatulas, small.* Two small wooden spatulas (tongue depressors) shall be packaged with each unit of paste resin and hardener as described in (e) above.

j. *Spatulas.* Spatulas shall be bundled and secured by tying or taping.

k. Brushes. Each brush shall be individually wrapped in kraft paper. The ends of the kraft paper shall be folded back over each end of the brush and secured with strips of water resistant, pressure sensitive adhesive tape. Brushes shall be arranged in a reversing end-to-end order and placed in a sung-fitting folding or set-up paperboard container as specified under (b) above.

l. *Paper, kraft.* The paper shall be folded to a minimum practical size and secured by taping or tying.

m. *Instructions.* One copy of instructions (see 3.3.17) shall be sealed in a waterproof transparent bag. Bag closure shall be effected by heat-sealing.

n. *Spreading tool.* Each spreading tool shall be enclosed in a waterproof, transparent, heat or cold sealed unit package. The unit package may be of the skin or blister type unit package construction.

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5.2.1.1 **Component parts.** The component parts, as required for each type kit (see 1.2), unit protected as specified in 5.2.1 shall be unit packaged as a kit within a fiberboard container as specified under (b) above. Containers shall be provided with separators, partitions, cells, or similar media, so as to provide a compact nonshifting load and to prevent damage to the content and container from movement within the kit container.

5.2.2 **Level C.** Component parts and the assembled kit shall be unit protected as specified under level A except that the paperboard and fiberboard containers shall be of the domestic or nonweather resistant class, variety, or process as applicable. Domestic and domestic-class fire-retardant fiberboard containers shall be closed in accordance with method I using pressure sensitive adhesive tape, and weather resistant fiberboard containers shall be closed in accordance with method V.

5.2.3 **Commercial.** Commercial packaging (cleaning, preservation, cushioning and the unit pack) of component parts and the assembled kit shall be in accordance with ASTM D 3951.

5.3 **Packing.** Packing of assembled kits, preserved as specified (see 5.2), shall be packed level A, B, C, or commercial as specified (see 6.2).

5.3.1 **General requirements for levels A, B, and C.** Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, contain identical quantities of identical kits, and be of uniform size.

5.3.2 **Levels A, B, and C containers.** Kits, preserved as specified (see 5.2), shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII, exterior shipping container requirements, of MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2), container selection and options shall be at the contractor's option.

5.3.2.1 **Caseliners, closure and gross weight.**

5.3.2.1.1 **Claseliners.** Unless otherwise specified (see 6.2), level A shipping containers containing kits preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 **Closure.** Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method B and reinforced with non-metallic or tape banding and domestic or fire-retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

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5.3.2.1.3 **Weight.** Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 **Commercial.** Kits preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 **Container modification.** Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment and storage.

5.4 **Marking, Levels A, B, C, and commercial.** In addition to any special marking required (see 3.5 and 6.2), interior (unit) packs and shipping containers shall be marked including bar coding shipment, stowage, and storage in accordance with MIL-STD-2073, appendix F.

5.5 **Material safety data sheet.** A copy of the material safety data sheet (see 3.7 and 6.7) shall be attached to the shipping document for each destination.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The type I kit is intended to provide a complete assemblage of materials (not tools) for making minor, permanent, structural repairs to glass reinforced plastic items, notably boats. The type II kit is a refill kit, particularly for users who do frequent repairs and already have the necessary stirring sticks, scissors, instruction manuals, and the like. The type III kit contains only those materials selected by the user activity.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Type of repair kit required (see 1.2)

d. When first article is required (see 3.1)

e. Materials, items, and quantities required for type III (see 3.3.1.3)

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f. Navy fire-retardant requirements (see 5.1.1)

g. Level of preservation and packing required (see 5.2 and 5.3)

h. Container selection and options (see 5.3.2)

i. When waterproof caseliners are required (see 5.3.2.1.1)

j. Special marking required (see 3.5 and 5.4).

6.3 **First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 **Sub-contracted material and parts.** The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 **Material disposal.** Disposal of the material components or the reacted compounds of those kits shall comply with applicable federal, state, and local regulations.

6.6 **Material safety data sheet (MSDS).** Contracting officers must identify those activities requiring copies of MSDSs. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.7 **Subject term (key word) listing.**

Epoxy
Resins
Thermosetting

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6.8 **Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army – ME
Navy – SH
Air Force – 82

Review activities:
Army – ME
Navy – SH
Air Force – 82

User Activities:
Navy – CG, AS, DS

Preparing activity:
Navy – SH
(Project 2090-0096)

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APPENDIX

INSTRUCTIONS FOR REPAIR OF REINFORCED PLASTICS

10. SCOPE

10.1 Scope. This appendix covers a brief instruction for use of a kit for repairing reinforced plastic. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. REFERENCED DOCUMENTS

20.1 Government documents.

20.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

DOD-P-15328	Primer (Wash), Pretreatment (Formula No. 117 for Metals); (Metric)
MIL-P-15931	Paint, Antifouling, Vinyl (Formulas No. 121 and No. 129)

20.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 0900-LP-006-0010	Reinforced Plastics Preventive Maintenance and Repair Manual
NAVSHIPS 0283-LP-101-7101	Ships Concealment Camouflage Instruction

(Unless otherwise indicated, copies of military specifications are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

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20.2 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

30. REQUIREMENTS

30.1 **Background information.** For further background information on repair of reinforced plastics, personnel shall refer to NAVSHIPS 0909-LP-006-0010.

30.2 **Introduction.** Damaged sections in reinforced plastic structures may be repaired by patching with materials similar to those used in the construction of the structure itself. The basic steps in repair of reinforced plastic are:

a. Assembling of necessary equipment (see 30.3)
b. Planning repair (see 30.4)
c. Preparation of damaged area (see 30.5)
d. Tailoring of reinforcement material (see 30.6)
e. Preparation of resin system (see 30.7)
f. Impregnation of reinforcement with resin (see 30.8 and 30.10)
g. Application of patch to hull (see 30.9 and 30.10)
h. Cure (hardening) of patch (see 30.11)
i. Finishing (see 30.12).

30.2.1 **Conditions.** The repair shall be accomplished in a sheltered area. The temperature shall not be below 50 °F.

30.2.2 **Safety.** Epoxy resins and hardeners have harmful vapors and can cause skin irritation, especially in sensitive individuals. Avoid exposure to vapor. Use respirator if adequate ventilation is not available. Avoid contact with the skin or eyes. Use protective clothing, gloves, and goggles. In case of accidental contact, promptly flush skin or eyes with plenty of water for 15 minutes; get medical attention for eyes.

CAUTION

Use only with liquid epoxy resin provided in this kit. (Add additional cautions/warnings as necessary for the specific materials involved.)

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APPENDIX

30.3 **Repair materials.** In addition to the repair materials supplied in the repair kit, some of the following tools and equipment will be needed for making the patch:

a. Chalk (for marking the area to be repaired)

b. Ruler

c. Scissors

d. Saw (metal cutting hand saw, hole or reciprocating saw, for cutting away damaged area)

e. Disc sander, cone sander, or file (for grinding away damaged portion and scarfing or beveling the edge of the cut out area)

f. Cardboard, sheet metal, or plywood panel (for use as a backing plate or cover)

g. Tape or bracing (for attaching or supporting the backing)

h. Acetone or lacquer thinner (for cleaning the surface and equipment)

i. Heat lamp (optional) (for speeding the cure of the resin)

j. Protective equipment such as goggles or a respirator.

30.4 **Planning the repair.** Outline the area to be cut away and the area to be scarfed (beveled) with chalk (see figures 6, 7, and 8). Determine the recommended length of scarf for the laminate thickness to be repaired, from scale B of figure 9. Round or oval shaped repairs are preferable. Do not use rectangular cuts with sharp corners. Estimate the amount of resin and glass reinforcement that will be required. To do this determine the thickness of the laminate and the average diameter of the patch, if the patch is round, or the average area, if the patch is rectangular or oval (see figure 10). Then turn to the Nomogram (figure 9) to obtain an estimate of the materials required. On the left hand side, see scale A: average diameter and average area. On the right hand side, see scale B: thickness. Draw a straight line between the average diameter or area and laminate thickness. Where this line intersects scale C, read the approximate amounts of cloth and resin required. The estimate is on the high side by 30 to 40 percent, particularly with respect to cloth, to allow for wastage. Scale B also indicates the approximate number of plies of cloth required for any given thickness.

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30.5 Preparing the damaged area. Surface gouges or small punctures may be repaired by using the paste resin in accordance with instructions on the resin tube. To repair more extensive damage, cut out the damage if the break is all the way through (see figures 7 and 8), or grind it away if it only extends partially through the laminate (see figure 6). Clean the adjacent area with acetone or lacquer thinner and grind the required scarf (bevel). After this has been done, apply a backing plate behind the hole. When accomplishing a repair on a vertical surface, a plate on each side of the patch is recommended to hold the patch in place while it is curing. The plates should be covered with separating film so that they may later be removed; they may be held in place by taping or shoring with lumber.

30.6 Tailoring the reinforcement. The glass reinforcement should be cut to fit the repair. The glass cloth should be used for general repairs. The glass mat may be used in fillets or for thicker build-ups, where strength is not of prime importance. Prepare paper templates (using the kraft paper) for the innermost ply, which will approximate the size of the hole, and the outermost ply, which should just overlap the scarfed area. Each of the intermediate plies (see figure 9 for the approximate number of plies required for a given thickness) should be cut proportionately larger than the preceding smaller ply.

30.7 Preparation of the resin system. Select the proper sized cans of resin ("A") and hardener ("B") (determine from figure 9 the quantity of resin required). In handling, observe the safety precautions indicated on the hardener can. Pour the hardener into the resin can and mix thoroughly. The resin "A" after mixing with the hardener "B" has a working life of approximately 1/2 hour at 73 °F. The resin will harden more quickly at higher temperatures and will take much longer to harden at lower temperatures. Cool or warm the sealed resin container in cold or hot water where necessary before mixing. Allow no water to come in contact with the contents.

30.8 Impregnating the glass reinforcing material with resin. The glass cloth may be laid up by the following procedure: Cut a piece of separating film larger than the largest tailored ply of glass cloth. The largest ply of the cloth is placed on the film and saturated with the resin-hardener mixture. The second largest tailored ply is centered over the first ply and impregnated. The procedure is continued, coating the successively smaller patch plies until all the plies of glass cloth are laid up, with the smallest ply on top. The resin should be apportioned so that there is enough for all the plies. The resin may be applied by brush, spatula, or spreader. Several plies of glass cloth may be saturated at one time. After all the glass cloth has been wetted, place another sheet of film on top of the wet cloth and work out the air bubbles by squeezing the layup from the center outward with a clean spatula or spreader.

30.9 Applying the patch. The patch shall be applied as follows: The entire sanded area around the hole is coated by brush with the same resin used for wetting the patch plies. Peel off the top layer of separating film (the one on top of the smallest ply of cloth). Leave the bottom layer of film in place to facilitate handling. The whole layup is now placed and centered over the

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hole with the separating film which was on the bottom, now on the outside. Leave this film in place. The patch is worked with the wooden spatula or spreading tool to remove any entrapped air. The worked patch should conform to the desired contour. The second (cover) plate which was previously prepared (see 30.5) is now placed over the separating film and clamped, taped, or braced in place so that the patch remains in firm contact with the original panel. The backing and cover plates are left in place from 8 to 16 hours to allow the patch to set up hard.

30.10 **Alternative method of impregnating the glass and applying the patch.** This procedure is useful for patches on a horizontal surface where the impregnated plies can be laid directly in place. The entire sanded area around the hole is coated by brush with resin-hardener mixture. The resin is also applied to the separating film on the backing plate. In this method the smallest ply is placed over the hole and impregnated with the resin. The next larger ply is then centered over the first ply and wetted with the resin. This procedure is continued until all the plies are in place with the largest ply on the top. In each case there should be sufficient resin so that it will squeeze through the next ply, pushing out entrapped air. A sheet of separating film is placed over the top ply and the entrapped air and excess resin is worked out from the center to the outside of the patch. A cover plate is clamped over the surface as described in 30.9 in order to maintain gentle pressure. A sandbag may be useful for covering the patch or applying pressure.

30.11 **Cure (hardening) of the patch.** The patch should remain undisturbed for at least 12 hours at 73 °F (longer at lower temperatures since the rate of curing approximately doubles for every 10 °C reduction in temperature). Heat lamps may be used to speed cure, but in using them care must be taken not to overheat the patched area, for if this is done the cure reaction may for too fast away, causing frothing, blisters, and porosity. Keep lamps at least 1-1/2 to 2 feet away so that the heated area is not too hot to comfortably touch.

30.12 **Finishing the patch.** After 12 to 24 hours at 73 °F, the patch should be sufficiently cured for finishing. Remove the cover plate and separating film. If the resin is not sufficiently hard, apply additional heat. The resin may soften slightly under heat if not well cured, but will harden when cured. Excess resin or other patch irregularities can be abraded away to make the patch fit the desired contour. If required, additional patching can be done over this patch after the surface has been sanded. After abrading to smoothness, apply one coat of pretreatment in accordance with DOD-P-15328 and vinyl alkyd paint as required by NAVSHIPS 0283-LP-101-7010 or, where required, antifoul formula number 121 in accordance with MIL-P-15931.

40. QUALITY ASSURANCE PROVISIONS

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APPENDIX

50. PACKAGING

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60. NOTES

60.1 Care of equipment. Equipment such as brushes, gloves, spreader, shears, and the like, used for preparation of the patch should be cleaned as soon as possible (before the resin hardens). The equipment can be washed with acetone or lacquer thinner (caution: highly inflammable solvents), or in strong soap powder, or other alkaline cleaning solutions.

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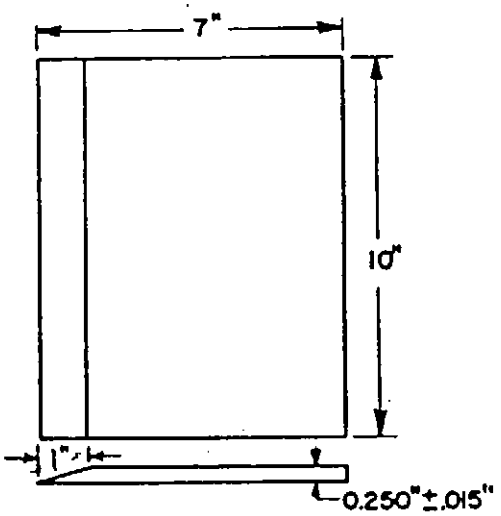


FIGURE 1. Standard panel.

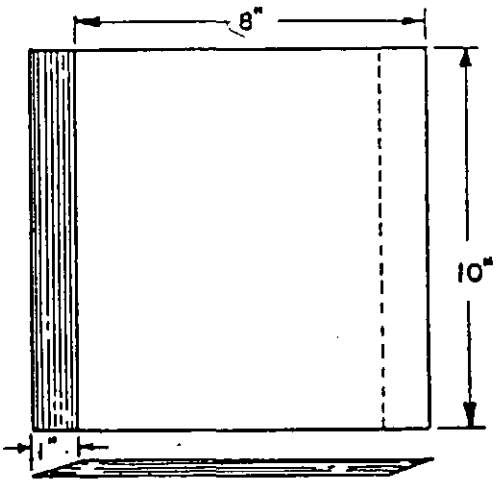


FIGURE 2. Standard patch.

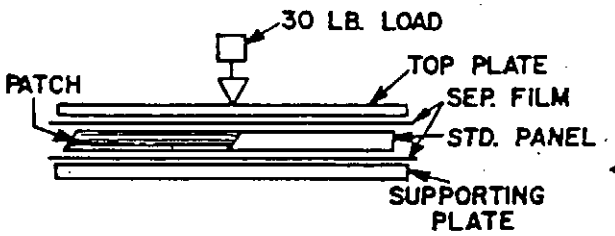


FIGURE 3. Application of patch.

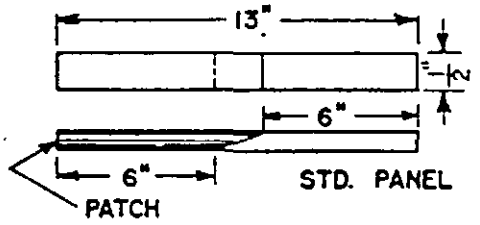
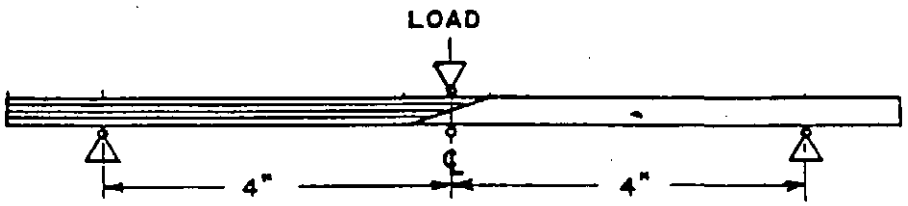


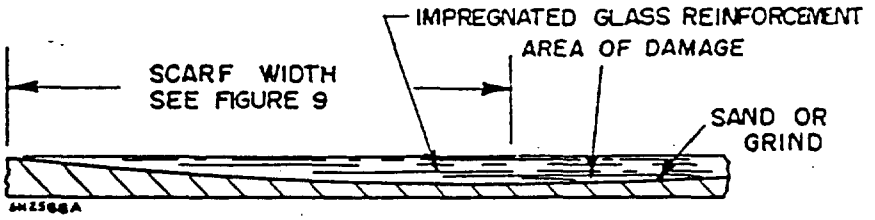
FIGURE 4. Specimen.



SH 333

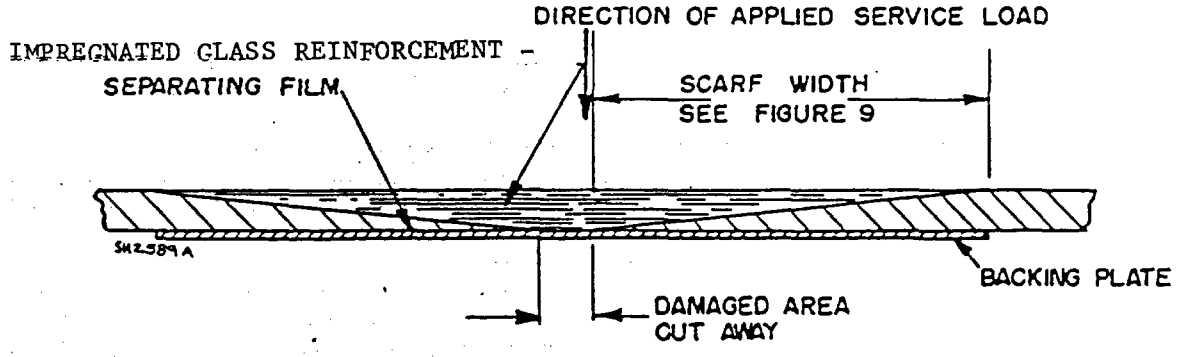
FIGURE 5. Flexural test of specimen.

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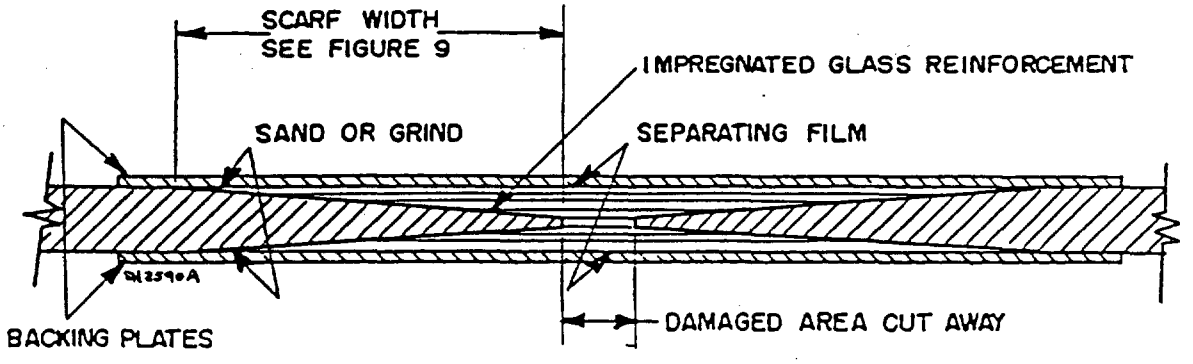
SH 2588A

FIGURE 6. Surface damage or damage through partial thickness of laminate.



SH 2589A

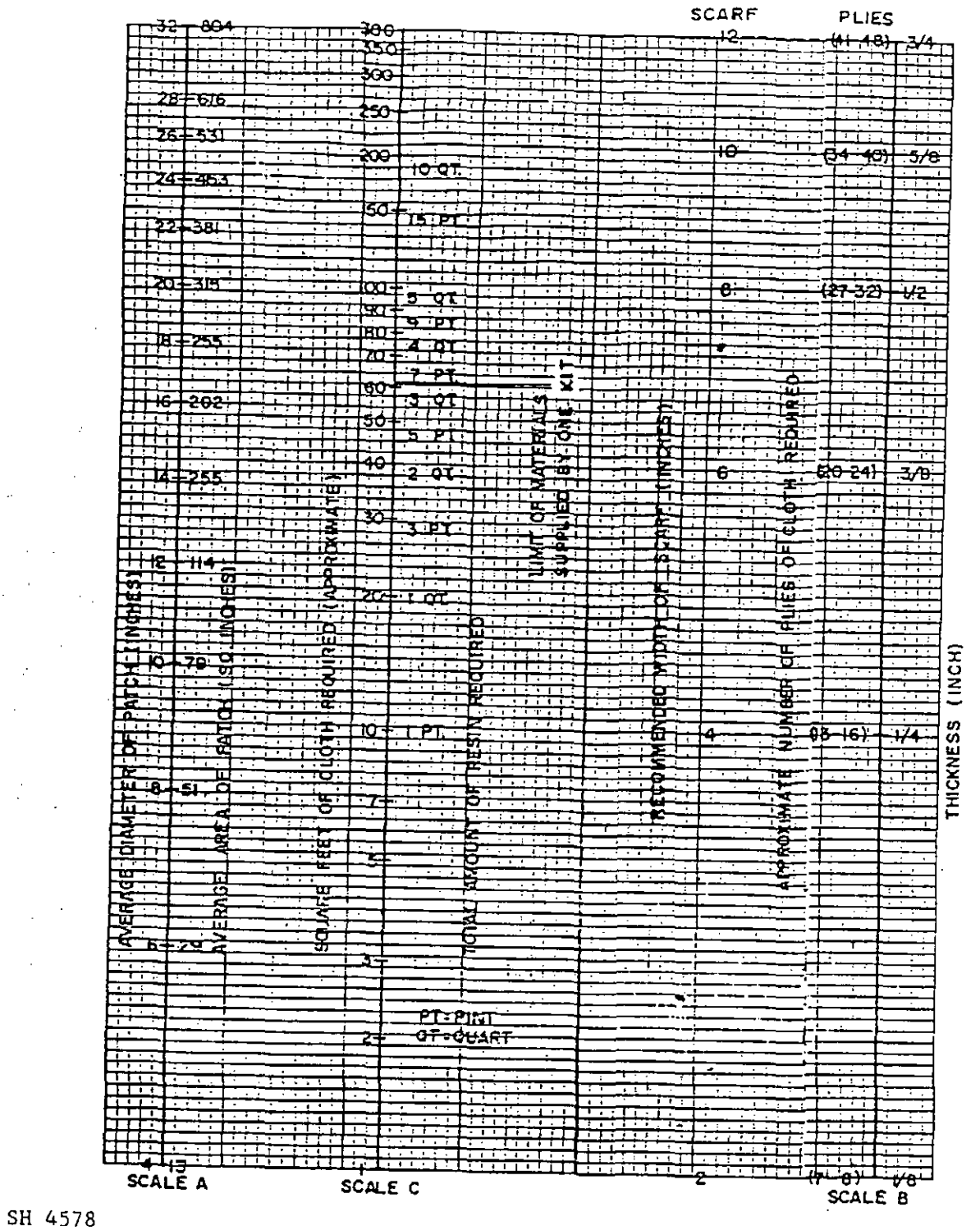
FIGURE 7. Damage through complete thickness of laminate; patch applied to one side (recommended where service loadings are light or are applied to one side only).



SH 2590A

FIGURE 8. Damage through complete thickness of laminate; patch applied from both sides (recommended where service load is severe or may be applied to both sides).

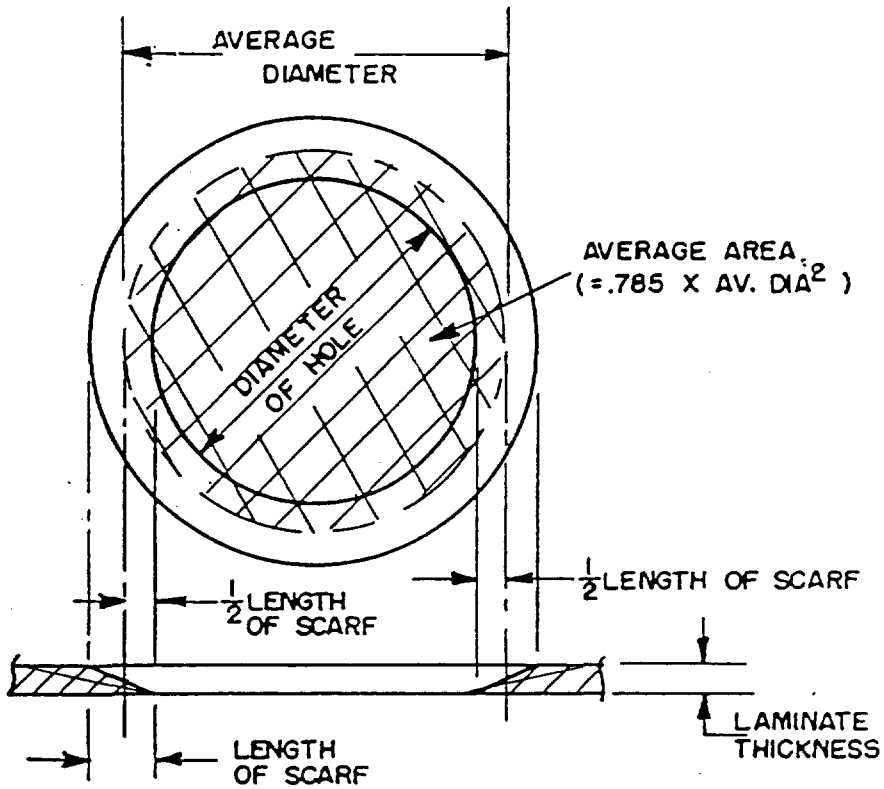
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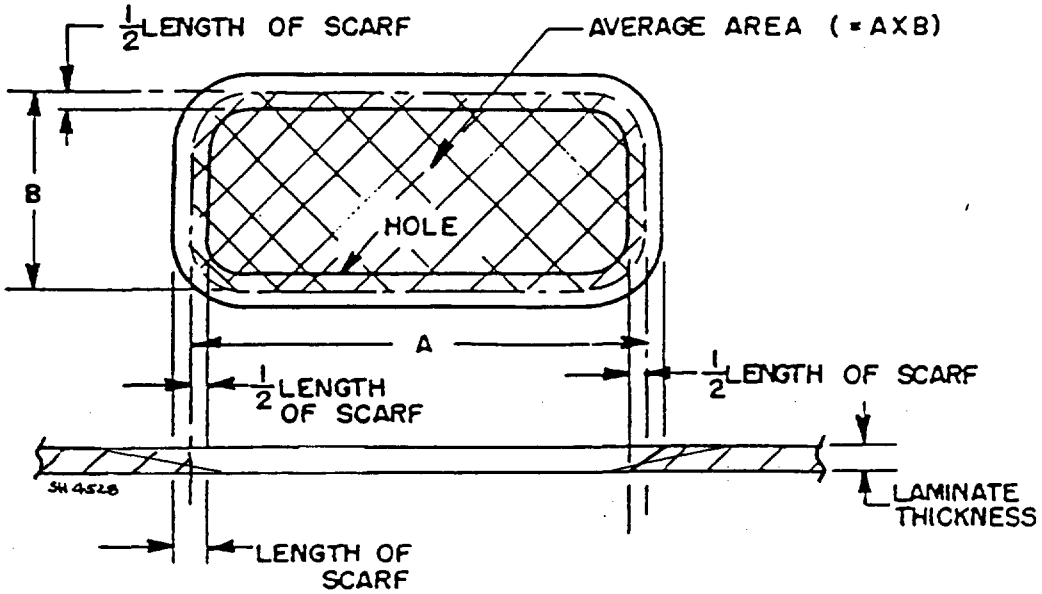
SH 4578

FIGURE 9. Estimating materials.

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CIRCULAR PATCH



RECTANGULAR PATCH

SH 4528

FIGURE 10. Average diameter and area of patch.

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